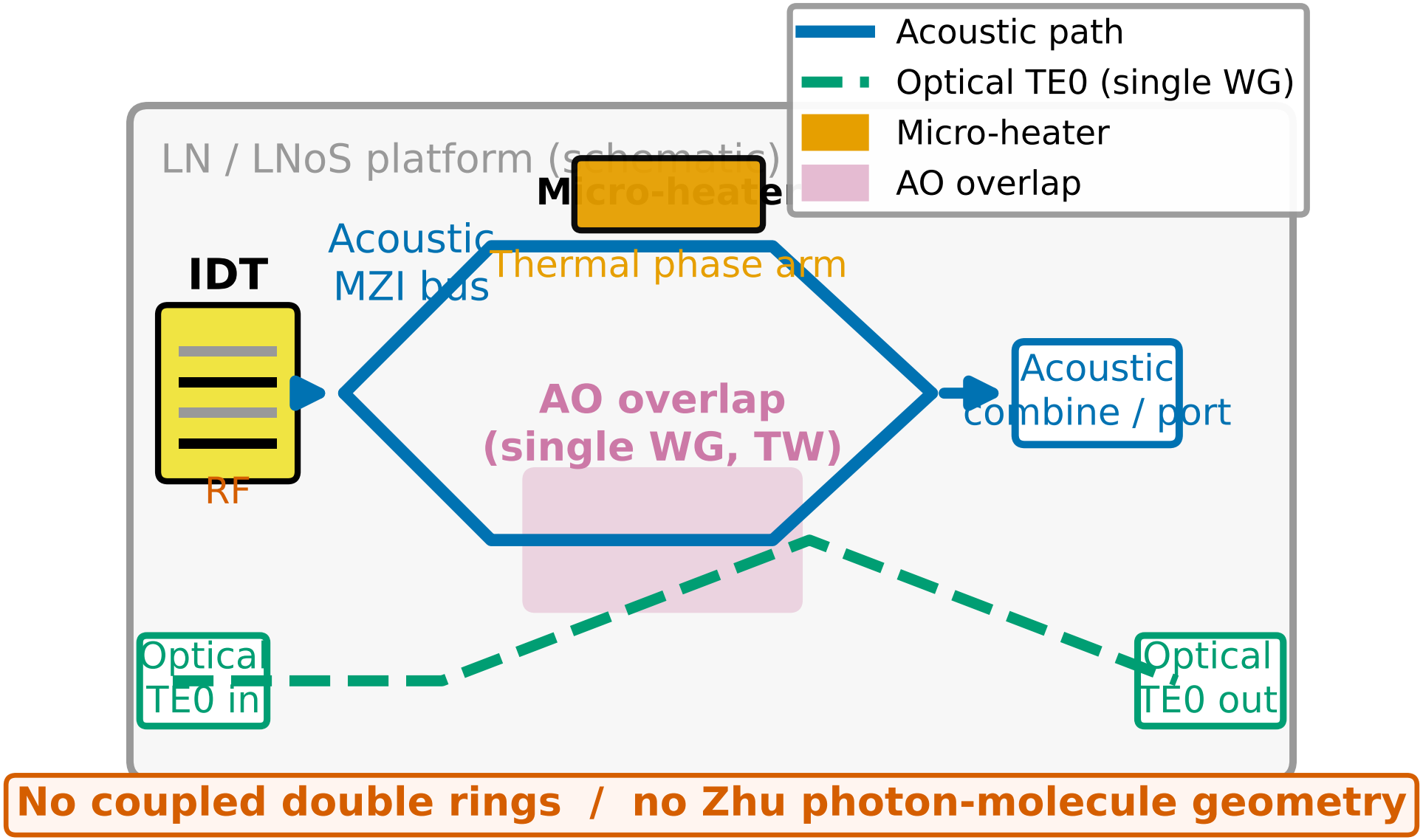Fig. 1 Device schematic (Direction 2)